Yizuo (Alex) Chen
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Portfolio

Education

M.Eng., Electrical and Computer Engineering, University of British Columbia *2026–2028*
B.Eng., Electrical Engineering, British Columbia Institute of Technology *2022–2026*
GPA: 89% (4.00/4.00)

Experience

Junior Control Engineer (Co-op), TKT Innovations Ltd., Langley, BC *May–Aug 2025*

- Optimized Siemens PLC logic for industrial dishwashing systems with Siemens TIA Portal, eliminating frequent plate drops and raw dog-food spills, improved line stability.
- Standardized control code architecture to align with industrial best practices and regulatory requirements.

Electrical Engineering Intern (Co-op), Canature Processing, Langley, BC *Jun–Sep 2023*

- Diagnosed and resolved production-line logic issues, reducing material waste and improving throughput.
- Executed test plans and authored QA documentation for certification readiness.

Lab Assistant, UBC AMPEL, Vancouver, BC *Jul–Sep 2022*

- Collaborated with graduate students to design a Fusion 360 enclosure for a research sample holder, enabling controlled rotation of samples under UV light.

Capstone Project: Social Robot for Pediatric Healthcare

BCIT & **UBC Faculty of Medicine** & **BC Children's Hospital** *Sep 2025–May 2026*
Researching and developing NAO robot features for pediatric anxiety relief, pain management, and rehabilitation support (BCIT News feature).

- Programmed NAO robot workflows in two modes: autonomous interactive support for pediatric anxiety and pain relief, and scripted programmable exercises for cerebral palsy rehabilitation.
- Built a Python/Vosk external speech-recognition script to enable flexible voice interaction beyond the robot's built-in speech-recognition tools.
- Programmed and recorded 18 pediatric rehabilitation exercises, integrating motion and speech routines for cerebral palsy recovery support.
- Developed a custom UI for UBC researchers to build exercise routines and a Python bridge connecting the Python 3 UI to the NAOqi SDK running on Python 2.7.

Projects

NFC Attendance System (RPi Pico 2W & FreeRTOS) — GitHub *Oct–Dec 2025*

- Built an NFC-based attendance system by porting and rewriting PN532 drivers for FreeRTOS on Pico 2W, enabling real-time UID logging with LED/buzzer feedback, onboard storage, and USB/Wi-Fi data offload.

Projects Continued

Face Recognition System (MATLAB & Digital Image Processing) — GitHub *Oct–Dec 2025*

- Built a real-time face recognition pipeline using RGB+HSV skin models and geometric priors, achieving robust multi-face detection with adaptive morphology and template correlation.

Ultrasonic Radar (DE0-Nano SoC & HC-SR04) — GitHub *Feb–Apr 2025*

- Built an ultrasonic radar by writing SystemVerilog for sensor timing, servo PWM, and UART, paired with a Python visualizer for automated sweeps and manual control.

Remote-Control Pi Car (CV & Targeting) *Apr–Jun 2024*

- Designed a custom PCB and enclosure and programmed C++/OpenCV control software, enabling both manual server–client driving and autonomous target detection on Raspberry Pi.

Automated Ball-Sorting Machine (C++ & OpenCV) — GitHub *Feb–Apr 2024*

- Built a vision-guided sorting system by integrating OpenCV with servo control, supporting automated, push-button, and remote modes for accurate queue-based sorting.

Portable DC Power Supply *Feb–Apr 2023*

- Designed and assembled a compact bench supply by creating the enclosure and PCB and validating a −25–25 V adjustable output for lab use.

Competitions & Activities

BCIT Engineering Competition – Senior Design *Second Place*

- Designed and built a slim, rigid frame for a desert-terrain competition vehicle, improving stability, protecting all electrical components, and ensuring reliable performance under harsh conditions.

Western Engineering Competition – Programming, University of Alberta — GitHub *Jan 2026*

- Competed as the Map Specialist for a real-time electric bus fleet simulation, implementing Python and JavaScript modules to parse route and charging-station data from CSV files and project E-bus lines onto an interactive map.
- Enabled live visualization of fleet movement, battery state, and charging behavior, supporting optimization algorithms that dynamically rerouted buses to charging stations under energy constraints.

Technical Skills

Programming & Tools: C, C++, Python, MATLAB/Simulink, SystemVerilog, OpenCV, VS Code, VS Studio, SolidWorks, LaTeX
Embedded & Interfaces: TI MCUs, Raspberry Pi, RPi Pico 2W, Arduino, UART/SPI/I2C, firmware development, troubleshooting & debugging
Hardware & PCB: Schematic capture, KiCad PCB layout, soldering & assembly, 3D printing (G-code)
Controls & DAQ: PLC programming (TIA Portal), automated system design, real-time data acquisition
Test & Measurement: Oscilloscope, DMM, Logic Analyzer, Spectrum Analyzer
Other: Assembly drawings, user manuals, documentation, technical reports, Class 5 Driver's License